GATE ME 90 Days Challenge

Day 58/90 · Practice Set with Detailed Solutions

SUBJECT	TODAY'S TOPIC	COUNTDOWN
Manufacturing	**Joining, Powder Metallurgy & Polymers**	**44 days to GATE**

How to use this set: 15 questions (8 MCQ, 7 NAT) on arc welding (polarity, power-source and arc characteristics, heat input, duty cycle), gas, resistance, thermit and solid-state welding, weld defects and carbon equivalent, brazing and soldering, powder production, compaction and sintering, and polymer and composite processing (moulding processes, rule of mixtures). Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.
Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 MCQ

Alternating current (AC) is generally preferred for TIG welding of aluminium because

(A) it gives the deepest penetration of all polarities

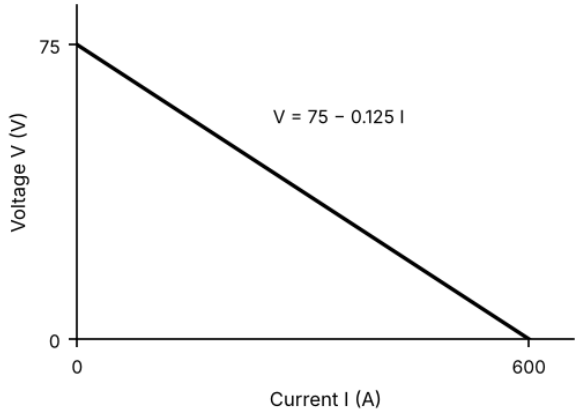
(B) the electrode-positive half-cycle breaks up the surface oxide film (cathodic cleaning) while the electrode-negative half-cycle keeps the tungsten cool

(C) aluminium cannot carry direct current

(D) AC needs no shielding gas

Q2 NAT

A welding power source has the linear characteristic shown (open-circuit voltage 75 V, short-circuit current 600 A). The arc characteristic is V = 20 + 5l, with l the arc length in mm. The arc length (in mm, round off to one decimal place) that gives the maximum arc power is _____



Q3 NAT

A butt weld is made at 30 V and 250 A with a travel speed of 10 mm/s. If 75 % of the arc energy enters the work, the net heat input per unit length (in J/mm, round off to one decimal place) is _____

Q4 MCQ

In a neutral oxy-acetylene flame, the oxygen needed for complete combustion of acetylene is supplied as follows:

(A) all of it from the oxygen cylinder

(B) 2.5 volumes from the cylinder and none from air

(C) 1.5 volumes from the cylinder and 1 volume from air

(D) 1 volume per volume of acetylene from the cylinder (inner cone) and 1.5 volumes from the surrounding air (outer envelope)

Q5 NAT

In a resistance spot weld, a current of 8 kA flows for 0.15 s through an effective resistance of 120 μΩ. The heat generated (in J) is _____

Q6 MCQ

In diffusion bonding (diffusion welding), the joint is formed by

(A) melting a thin filler layer between the parts

(B) a high-velocity impact that produces a wavy interface

(C) atomic diffusion across clean, closely mated surfaces held under pressure at an elevated temperature, without melting

(D) frictional heating produced by relative rotation

Q7 MCQ

In brazing, the joint clearance is kept small (typically 0.025–0.25 mm) mainly because

(A) the filler metal is drawn into and held in the joint by capillary action

(B) a small gap needs no flux

(C) the base metal melts only in narrow gaps

(D) a large clearance makes the joint too strong

Q8 NAT

A welding transformer is rated at 250 A for a 60 % duty cycle. The duty cycle at which it may be used at 200 A (in %, round off to two decimal places) is _____

Q9 MCQ

Which one of the following products is a typical application where powder metallurgy is clearly superior to other processes?

(A) Large crankshafts

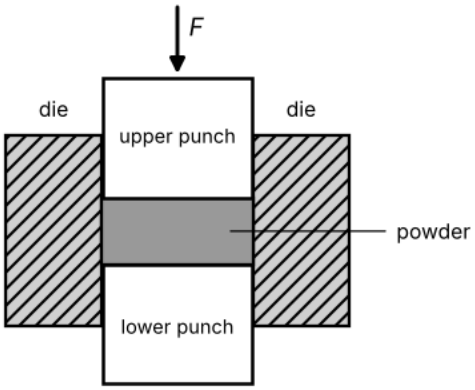
(B) Thin-walled aluminium beverage cans

(C) Long structural I-beams

(D) Self-lubricating porous bronze bearings

Q10 NAT

A gear blank with a projected area of 1500 mm² is compacted in a die (see figure) at a pressure of 600 MPa. The press force needed (in kN) is _____



Q11 NAT

An iron powder compact has a green density of 6.4 g/cm³. The theoretical density of iron is 7.87 g/cm³. The porosity of the compact (in %, round off to one decimal place) is _____

Q12 MCQ

In the atomization of molten metals to make powders,

(A) water atomization gives irregular particles and gas atomization gives nearly spherical particles

(B) both water and gas atomization always give flake-shaped particles

(C) gas atomization gives irregular particles with the best green strength

(D) atomization can be used only for brittle metals

Q13 MCQ

Which one of the following polymers can be softened by heating and re-shaped several times?

(A) Bakelite (phenol formaldehyde)

(B) Cured epoxy resin

(C) Polyethylene

(D) Vulcanized rubber

Q14 NAT

A unidirectional glass-fibre/epoxy composite has 60 % fibres by volume (E_f = 72 GPa, E_m = 3 GPa). Its elastic modulus transverse to the fibres (in GPa, round off to two decimal places) is _____

Q15 MCQ

Pultrusion is used to produce

(A) continuous fibre-reinforced polymer sections of constant cross-section, such as rods, channels and ladder rails

(B) hollow plastic bottles

(C) thin polymer films

(D) large hollow seamless tanks

Answer Key

Q	Type	Answer
Q1	MCQ	(B) the electrode-positive half-cycle breaks up the surface oxide film (cathodic cleaning) while the electrode-negative half-cycle keeps the tungsten cool
Q2	NAT	3.5 (accept 3.4 to 3.6)
Q3	NAT	562.5 (accept 559.7 to 565.3)
Q4	MCQ	(D) 1 volume per volume of acetylene from the cylinder (inner cone) and 1.5 volumes from the surrounding air (outer envelope)
Q5	NAT	1152
Q6	MCQ	(C) atomic diffusion across clean, closely mated surfaces held under pressure at an elevated temperature, without melting
Q7	MCQ	(A) the filler metal is drawn into and held in the joint by capillary action
Q8	NAT	93.75 (accept 93.30 to 94.20)
Q9	MCQ	(D) Self-lubricating porous bronze bearings
Q10	NAT	900
Q11	NAT	18.7 (accept 18.5 to 18.9)
Q12	MCQ	(A) water atomization gives irregular particles and gas atomization gives nearly spherical particles
Q13	MCQ	(C) Polyethylene
Q14	NAT	7.06 (accept 7.00 to 7.12)
Q15	MCQ	(A) continuous fibre-reinforced polymer sections of constant cross-section, such as rods, channels and ladder rails

Detailed Solutions

Q1 Answer: (B) the electrode-positive half-cycle breaks up the surface oxide film (cathodic cleaning) while the electrode-negative half-cycle keeps the tungsten cool

Concept: Aluminium is covered by a tough, high-melting Al_2O_3 film that must be removed for the weld pool to fuse.

When the electrode is positive, positive ions bombard the work (cathode) and scrub off the oxide; when it is negative, most heat goes into the work and the tungsten stays cool.

AC combines both effects: **(B)**.

Exam tip: DCEN is used for TIG welding of steels and stainless steels; DCEP would overheat a tungsten electrode.

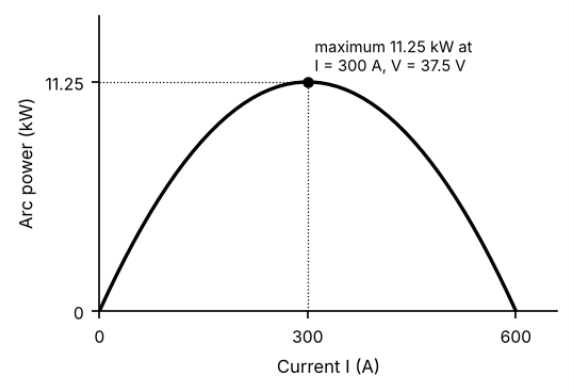
Q2 **Answer: 3.5 (accept 3.4 to 3.6)**

Concept: On a linear source $V = V_0(1 - I/I_s)$, the power VI is maximum at $V = V_0/2$ and $I = I_s/2$.

$P = I(75 - 0.125I)$; $dP/dI = 75 - 0.25I = 0 \rightarrow I = 300$ A, $V = 37.5$ V, $P_{max} = 11.25$ kW (see figure).

Arc: $37.5 = 20 + 5l \rightarrow l =$ **3.5 mm**

Exam tip: $P_{max} = V_0 I_s/4$ is a quick check: $75 \times 600/4 = 11\,250$ W.



Q3 **Answer: 562.5 (accept 559.7 to 565.3)**

Concept: Net heat input $H = \eta VI/v$ (energy delivered per mm of weld).

$VI = 30 \times 250 = 7500$ W

$H = 0.75 \times 7500/10 =$ **562.5 J/mm**

Exam tip: Higher heat input lowers the cooling rate, which reduces HAZ hardness but coarsens the grains.

Q4 **Answer: (D) 1 volume per volume of acetylene from the cylinder (inner cone) and 1.5 volumes from the surrounding air (outer envelope)**

Concept: Complete combustion: $C_2H_2 + 2.5\ O_2 \rightarrow 2CO_2 + H_2O$.

Primary reaction in the inner cone: $C_2H_2 + O_2 \rightarrow 2CO + H_2$ (equal volumes from the cylinders; hottest point ≈ 3200 °C just beyond the cone).

Secondary reaction in the outer envelope: $2CO + H_2 + 1.5\ O_2 \rightarrow 2CO_2 + H_2O$, with oxygen from the air: **(D)**.

Exam tip: Because the inner zone contains CO and H_2, the neutral flame shields the pool from oxidation.

Q5 **Answer: 1152**

Concept: Joule heating $H = I^2Rt$.

$I^2 = (8000)^2 = 6.4 \times 10^7$ A²; $R = 1.2 \times 10^{-4}$ Ω; $t = 0.15$ s

$H = 6.4 \times 10^7 \times 1.2 \times 10^{-4} \times 0.15 =$ **1152 J**

Exam tip: The contact resistance between the sheets is the largest part of R, which is why the nugget forms at the faying surface.

Q6 Answer: (C) atomic diffusion across clean, closely mated surfaces held under pressure at an elevated temperature, without melting

Concept: Solid-state processes join metals without melting the base metal.

In diffusion bonding the surfaces are pressed together at about 0.5–0.8 T_m for minutes to hours; atoms diffuse across the interface and voids close up: **(C)**.

(B) describes explosion welding and (D) friction welding.

Exam tip: Diffusion bonding is used for titanium aerospace parts and for joining dissimilar metals and ceramics.

Q7 Answer: (A) the filler metal is drawn into and held in the joint by capillary action

Concept: Brazing depends on wetting and capillary flow of the molten filler between closely fitting surfaces.

If the gap is too large, capillary forces cannot pull the filler in and the joint strength falls towards that of the filler; too small a gap stops the flow: **(A)**.

A properly brazed thin joint can be stronger than the filler alone, because the joint is constrained by the stronger base metal.

Exam tip: Flux is still needed in most brazing to remove oxides and help wetting.

Q8 Answer: 93.75 (accept 93.30 to 94.20)

Concept: The heating of the source $\propto I^2 \times$ duty cycle, so I^2D is constant.

$D_2 = D_1(I_1/I_2)^2 = 60 \times (250/200)^2 = 60 \times 1.5625$

$D_2 =$ **93.75 %**

Exam tip: The current that may be drawn continuously (100 %) is $250 \times \sqrt{0.6} = 193.6$ A.

Q9 Answer: (D) Self-lubricating porous bronze bearings

Concept: PM can deliberately leave a controlled, interconnected porosity.

Sintered bronze bearings are vacuum-impregnated with oil (20–30 % porosity) and lubricate themselves in service: **(D)**.

Crankshafts are forged, cans are deep drawn and ironed, and I-beams are rolled.

Exam tip: Other PM-only products: cemented carbide tools, tungsten filaments and porous metal filters.

Q10 Answer: 900

Concept: Compaction force = compaction pressure × projected area of the part normal to the pressing direction.

$F = 600\ \text{N/mm}^2 \times 1500\ \text{mm}^2 = 9.0 \times 10^5$ N

$F =$ **900 kN**

Exam tip: In single-action pressing the density falls with distance from the moving punch; double-action pressing gives a more uniform density.

Q11 **Answer: 18.7 (accept 18.5 to 18.9)**

Concept: Porosity = 1 − (actual density/theoretical density).

Relative density = 6.4/7.87 = 0.8132

Porosity = 1 − 0.8132 = 0.1868 = **18.7 %**

Exam tip: Sintering and re-pressing raise the relative density; most structural PM steels are 85–92 % dense.

Q12 **Answer: (A) water atomization gives irregular particles and gas atomization gives nearly spherical particles**

Concept: Particle shape depends on how fast the droplets freeze compared with how fast surface tension rounds them.

Water quenches droplets very quickly, freezing them in irregular shapes (good interlocking, good green strength); in inert gas they have time to spheroidize: **(A)**.

Spherical powders flow and pack well (used in metal injection moulding and additive manufacturing).

Exam tip: Brittle materials are powdered by crushing/milling; flake powders come from ball milling ductile metals.

Q13 **Answer: (C) Polyethylene**

Concept: Thermoplastics have linear or branched chains held by secondary bonds that weaken on heating; thermosets and vulcanized rubbers are cross-linked by primary bonds.

Polyethylene is a thermoplastic and can be re-melted and recycled: **(C)**.

Bakelite, cured epoxy and vulcanized rubber decompose rather than melt on heating.

Exam tip: Thermoplastics → injection moulding, extrusion, blow moulding; thermosets → compression and transfer moulding.

Q14 **Answer: 7.06 (accept 7.00 to 7.12)**

Concept: Loaded across the fibres, the phases carry the same stress (iso-stress): $1/E_t = V_f/E_f + V_m/E_m$.

$1/E_t = 0.6/72 + 0.4/3 = 0.00833 + 0.13333 = 0.14167\ GPa^{-1}$

E_t = **7.06 GPa**

Exam tip: Along the fibres (iso-strain) E = 0.6 × 72 + 0.4 × 3 = 44.4 GPa; the transverse value is governed by the soft matrix.

Q15 **Answer: (A) continuous fibre-reinforced polymer sections of constant cross-section, such as rods, channels and ladder rails**

Concept: In pultrusion, continuous fibre rovings are pulled through a resin bath and then through a heated die that shapes and cures the section.

So it makes constant-section, fibre-reinforced profiles of any length: **(A)**.

Bottles are blow moulded, films are blown, tanks are rotationally moulded or filament wound.

Exam tip: Pultrusion is the composite counterpart of extrusion, but the material is *pulled*, not pushed, through the die.

What next?

Day 58 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

Website test series

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Telegram: daily PDFs

Next, Day 59/90: Manufacturing, Machining Mechanics & Tool Life. Comment **GATE90** on our reel to get all links.
Want concise revision notes? The GATE ME handwritten short notes of an AIR 16 topper are available in the Dig Career Thrust app.